Worksheet 1.3 — Sigma Notation

Mathematics: Analysis & Approaches SL (IB) · Unit 1: Number and Algebra

The Big Ideas

Sigma notation $\sum_{n=1}^{N} f(n)$ compactly writes the sum of $f(n)$ from $n = 1$ to $n = N$.

- $\sum_{n=a}^{b} f(n)$: substitute each integer n from a to b into $f(n)$ and add.
- If $f(n)$ is linear in n, the sum is an arithmetic series; if $f(n)$ involves r^n, it's geometric.
- Splitting or shifting the index (e.g. starting at $n = 0$ vs $n = 1$) can change the formula without changing the sum's value.

Worked Examples

Example 1 — Evaluate directly

Evaluate $\sum_{n=1}^{5}(3n - 2)$.

Solution. Terms: $1, 4, 7, 10, 13$.

Conclusion: sum $= 35$.

Example 2 — Geometric sigma sum

Evaluate $\sum_{n=1}^{6} 4(2)^{n-1}$.

Solution. This is a geometric series with $u_1 = 4$, $r = 2$, $n = 6$ terms: $S_6 = \dfrac{4(2^6 - 1)}{2 - 1} = 4(63)$.

Conclusion: 252.

Example 3 — Write a series in sigma form

Write $8 + 13 + 18 + \cdots + 68$ in sigma notation and evaluate it.

Solution. **Step 1:** Arithmetic, $u_1 = 8$, $d = 5$, general term $u_n = 8 + 5(n - 1) = 5n + 3$.

Step 2: Find n: $5n + 3 = 68 \Rightarrow n = 13$. So $\sum_{n=1}^{13}(5n + 3)$.

Step 3: $S_{13} = \dfrac{13}{2}(8 + 68) = \dfrac{13}{2}(76)$.

Conclusion: sigma form $\sum_{n=1}^{13}(5n + 3)$, value $= 494$.

Example 4 — Application — total savings

A savings plan deposits $D_n = 25 + 3n$ dollars in month n, for $n = 1$ to 12. Find the total deposited over the year.

Solution. $\displaystyle\sum_{n=1}^{12}(25+3n) = 12(25) + 3\sum_{n=1}^{12} n = 300 + 3\left(\frac{12(13)}{2}\right) = 300 + 3(78).$

Conclusion: total $= \$534$.

Example 5 — Splitting a sum into known pieces

Evaluate $\displaystyle\sum_{n=1}^{10}(4n^2 - 3n)$ using $\sum n = \frac{n(n+1)}{2}$ and $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$.

Solution. $\sum_{n=1}^{10} n^2 = \frac{10(11)(21)}{6} = 385$; $\sum_{n=1}^{10} n = 55$.

Conclusion: $4(385) - 3(55) = 1540 - 165 = 1375$.

Example 6 — Shifted-index geometric sum

Evaluate $\displaystyle\sum_{n=0}^{5} 3(2)^n$.

Solution. This is geometric with $u_1 = 3$ (at $n = 0$), $r = 2$, and 6 terms (from $n = 0$ to 5): $S_6 = \dfrac{3(2^6 - 1)}{1}$.

Conclusion: $S_6 = 3(63) = 189$.

Example 7 — Application — attendance decline

A festival's attendance is modelled by $A_n = 2000 - 40n$ for day $n = 1$ to 10. Find the total attendance over the 10 days.

Solution. $\displaystyle\sum_{n=1}^{10}(2000 - 40n) = 10(2000) - 40\sum_{n=1}^{10} n = 20000 - 40(55).$

Conclusion: total $= 20000 - 2200 = 17800$ attendees.

Example 8 — Find an unknown upper limit

Find N such that $\displaystyle\sum_{n=1}^{N}(2n - 1) = 144$.

Solution. **Step 1:** This is arithmetic ($u_1 = 1, d = 2$): $S_N = \dfrac{N}{2}(2 + 2(N-1)) = N^2$.

Conclusion: $N^2 = 144 \Rightarrow N = 12$.

Example 9 — Sigma sum requiring factoring first

Evaluate $\displaystyle\sum_{n=1}^{5} n(n+1)$ by expanding first.

Solution. $n(n+1) = n^2 + n$, so the sum is $\sum n^2 + \sum n = \frac{5(6)(11)}{6} + \frac{5(6)}{2} = 55 + 15$.

Conclusion: 70.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Evaluate $\sum_{n=1}^{7}(2n+1)$.

Question 2

Evaluate $\sum_{n=1}^{5} 5(3)^{n-1}$.

Question 3

Evaluate $\sum_{k=3}^{6}(k^2-1)$.

Question 4

Write $4+9+14+\cdots+64$ in sigma notation and evaluate it.

Question 5

Write $2 + 6 + 18 + \cdots + 486$ in sigma notation and evaluate it.

Question 6

Evaluate $\sum_{n=0}^{4} 7(0.5)^n$.

Question 7

A theatre adds $D_n = 40 + 2n$ extra dollars in ticket sales on day n of a promotion, for $n = 1$ to 10. Find the total extra revenue.

Question 8

Evaluate $\sum_{n=1}^{8} n^2$ using $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$.

Question 9

Evaluate $\sum_{n=1}^{6} n(n-1)$ by expanding first.

Question 10

Find N such that $\sum_{n=1}^{N}(2n-1)=225$.

Question 11 — Challenge

Show that $\sum_{n=1}^{5}(2n+3)$ and $\sum_{n=0}^{4}(2n+5)$ give the same total, and explain why.

Question 12 — Challenge

Evaluate $\sum_{n=1}^{12}(3n^2-2n)$ using the standard sum formulas.

Question 13 — Challenge

A model predicts sales of $S_n = 500 - 15n$ units in week n, for $n = 1$ to 15. Find total sales over the 15 weeks.

Question Recap & Answer Key

Question Recap

1. Evaluate $\sum_{n=1}^{7}(2n+1)$.
2. Evaluate $\sum_{n=1}^{5} 5(3)^{n-1}$.
3. Evaluate $\sum_{k=3}^{6}(k^2-1)$.
4. Write $4+9+14+\cdots+64$ in sigma notation and evaluate it.
5. Write $2+6+18+\cdots+486$ in sigma notation and evaluate it.
6. Evaluate $\sum_{n=0}^{4} 7(0.5)^n$.
7. A theatre adds $D_n = 40+2n$ extra dollars in ticket sales on day n of a promotion, for $n=1$ to 10. Find the total extra revenue.
8. Evaluate $\sum_{n=1}^{8} n^2$ using $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$.
9. Evaluate $\sum_{n=1}^{6} n(n-1)$ by expanding first.
10. Find N such that $\sum_{n=1}^{N}(2n-1) = 225$.
11. Show that $\sum_{n=1}^{5}(2n+3)$ and $\sum_{n=0}^{4}(2n+5)$ give the same total, and explain why.
12. Evaluate $\sum_{n=1}^{12}(3n^2-2n)$ using the standard sum formulas.
13. A model predicts sales of $S_n = 500-15n$ units in week n, for $n=1$ to 15. Find total sales over the 15 weeks.

✓ Answer Key

1. terms $3,5,\ldots,15$: sum $= 63$
2. $S_5 = \frac{5(3^5-1)}{2} = 605$
3. $8+15+24+35 = 82$
4. $u_n = 5n-1$, $n=13$; $\sum_{n=1}^{13}(5n-1)$, $S_{13} = \frac{13}{2}(4+64) = 442$
5. $u_n = 2(3)^{n-1}$, $n=6$; $\sum_{n=1}^{6} 2(3)^{n-1}$, $S_6 = \frac{2(3^6-1)}{2} = 728$
6. $S = \frac{7(1-0.5^5)}{1-0.5} = 13.5625$
7. $\sum_{n=1}^{10}(40+2n) = 400+2(55) = 510$
8. $\frac{8(9)(17)}{6} = 204$
9. $\sum n^2 - \sum n = \frac{6(7)(13)}{6} - \frac{6(7)}{2} = 91-21 = 70$
10. $N^2 = 225 \Rightarrow N = 15$
11. Both give $5+7+9+11+13 = 45$ — shifting the index start from 1 to 0 while adjusting the formula's constant leaves every actual term unchanged
12. $3\sum n^2 - 2\sum n = 3\left(\frac{12(13)(25)}{6}\right) - 2\left(\frac{12(13)}{2}\right) = 3(650)-2(78) = 1950-156 = 1794$
13. $15(500) - 15\sum_{n=1}^{15} n = 7500 - 15(120) = 7500 - 1800 = 5700$ units